

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : mechanical Engineering Semester :VI Subject Code & Name: Renewable energy resources- BTMEC605C Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Explain in detail energy reserves in India	CO1	6
B)	Explain attenuation of solar radiation in Earth's atmosphere	CO1	6
C)	Differential between Renewable Energy sources and non-Renewable Energy sources.	CO1	6
Q.2	Solve Any Two of the following.		
A)	Explain with neat sketch construction and working of pyroheliometer	CO 2	6
B)	Define a) Declination angle b) Solar Azimuth angle c) Altitude angle.	CO2	6
C)	Calculate angle made by beam radiation with normal to a flat plate collector on May 1st 09. 00 h. The collector is located in New delhi (28° 35'N, 77° 12'E). It is tilted at an angle of 36° with the horizontal and is pointing due south, also calculate for 12.00h (Local apparent time).	CO 2	6
Q. 3	Solve Any two of the following.		
A)	Explain any two types of concentration collectors with neat sketch.	CO 2	6
B)	Explain construction and selection criteria for flat plate collector.	CO 2	6
C)	Explain testing of flat plate collector with neat sketch.	CO 2	6
Q.4	Solve Any Two of the following.		
A)	Explain construction and working of Photo Voltaic cell.	CO 3	6
B)	What is wind energy? Explain selection of site for wind turbine generation system.	CO 4	6
C)	Explain construction and working of Photo Voltaic cell.	CO 3	6
Q. 5	Solve Any Two of the following.		
A)	What is OTEC system? Explain working of open cycle OTEC system.	CO 4	6
B)	Explain Nuclear Reactor	CO 4	6
C)	What is biomass energy? Explain with neat sketch floating drum type	CO 4	6

	Biogas plant.		
	*** End ***		

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